

Core Entity & ERD Design v2.0

Organization Management Platform

Status: FROZEN

1. Architecture Foundation

Architecture Style:

- Modular Monolith
- Domain Driven Design (DDD)
- Event Driven Core

Principles:

- Permission-based visibility
 - Soft deletes
 - Audit-first design
 - Event-driven analytics
 - Extensible domain boundaries
 - No hierarchy-based access control
-

2. Root Organization Domain

Organization

Fields

- id
- name
- code
- status
- created_at
- updated_at

Relationships

- Organization → Departments
- Organization → Teams
- Organization → Users

- Organization → Roles
 - Organization → Permission Groups
 - Organization → Dashboards
 - Organization → Integrations
-

3. Identity Domain

User

Fields

- id
- employee_code
- first_name
- last_name
- email
- phone
- designation
- profile_photo
- joining_date
- status
- created_at
- updated_at
- deleted_at

Relationships

- User ↔ DepartmentMember
 - User ↔ TeamMember
 - User ↔ ProjectMember
 - User ↔ TaskAssignee
 - User ↔ SubtaskAssignee
 - User ↔ UserManager
 - User ↔ Roles
 - User ↔ Permission Groups
-

UserManager

Supports multiple managers.

Fields

- id

- user_id
 - manager_id
 - created_at
-

4. Organization Structure Domain

Department

Fields

- id
 - organization_id
 - name
 - description
 - status
-

DepartmentMember

Fields

- id
 - department_id
 - user_id
 - role_in_department
 - joined_at
-

Team

Fields

- id
 - organization_id
 - name
 - description
 - status
-

TeamMember

Fields

- id
 - team_id
 - user_id
 - role_in_team
 - joined_at
-

5. Security Domain

Role

Fields

- id
 - organization_id
 - name
 - description
-

Permission

Fields

- id
- code
- name
- description

Examples

- project.create
 - task.edit
 - analytics.view.department
-

RolePermission

Fields

- id

- role_id
 - permission_id
-

UserPermission

Fields

- id
 - user_id
 - permission_id
 - granted_by
 - granted_at
-

PermissionGroup

Fields

- id
- organization_id
- name
- description
- is_system_group

Examples

- HR Team
 - Project Managers
 - Department Heads
-

PermissionGroupMember

Fields

- id
 - group_id
 - user_id
-

PermissionGroupPermission

Fields

- id
 - group_id
 - permission_id
-

PermissionOverride

Fields

- id
- user_id
- permission_id
- effect
- created_by
- created_at

Values

- ALLOW
- DENY

Permission Resolution Order

1. DENY Override
 2. ALLOW Override
 3. Direct User Permissions
 4. Permission Group Permissions
 5. Role Permissions
 6. DENY
-

6. Project Domain

Project

Fields

- id
- code
- title
- description

- project_phase
- status
- priority
- start_date
- due_date
- completion_percentage
- workflow_id
- created_by
- created_at
- updated_at
- deleted_at

Project Phases

- Idea
- Planning
- Active
- On Hold
- Completed
- Archived
- Cancelled

ProjectMember

Fields

- id
- project_id
- user_id
- project_role
- joined_at
- is_active

ProjectDepartment

Fields

- id
- project_id
- department_id

ProjectTeam

Fields

- id
 - project_id
 - team_id
-

7. Task Domain

Task

Fields

- id
- title
- description
- status
- priority
- start_date
- due_date
- estimated_hours
- actual_hours
- workflow_id
- created_by
- created_at
- updated_at
- deleted_at

Relationships

- Many Projects
 - Many Assignees
 - Many Documents
 - Many Tags
 - Many Comments
-

ProjectTask

Fields

- id
- project_id

- task_id

Supports task belonging to multiple projects.

TaskAssignee

Fields

- id
 - task_id
 - user_id
-

Subtask

Fields

- id
 - task_id
 - title
 - description
 - status
 - priority
 - due_date
-

SubtaskAssignee

Fields

- id
 - subtask_id
 - user_id
-

TaskDependency

Fields

- id
- predecessor_task_id
- successor_task_id

Supports:

- Gantt Charts
 - Critical Path
-

Checklist

Fields

- id
 - task_id
 - title
 - completed
-

8. Generic Comment Domain

Comment

Fields

- id
- entity_type
- entity_id
- user_id
- content
- created_at

Can attach to:

- Project
 - Task
 - Document
 - Approval
 - Workflow
-

9. Workflow Domain

Workflow

Fields

- id
- name
- type

Types

- Global
 - Project Specific
-

WorkflowStatus

Fields

- id
 - workflow_id
 - name
 - sequence
-

WorkflowTransition

Fields

- id
 - workflow_id
 - from_status_id
 - to_status_id
 - conditions
-

10. Approval Domain

Approval

Fields

- id
 - entity_type
 - entity_id
 - status
 - requested_by
 - created_at
-

ApprovalAction

Fields

- id
 - approval_id
 - user_id
 - action
 - comments
 - timestamp
-

11. Attendance Domain

Attendance

Fields

- id
 - user_id
 - date
 - check_in
 - check_out
 - total_hours
 - status
-

LeaveRequest

Fields

- id
- user_id
- leave_type
- start_date
- end_date
- reason
- status

Approved By

- Admin
 - HR
-

12. Time Tracking Domain

Timesheet

Fields

- id
 - user_id
 - task_id
 - date
 - hours_logged
 - notes
-

13. Document Domain

Folder

Fields

- id
- parent_folder_id
- name

Supports nested folders.

Document

Fields

- id
 - folder_id
 - name
 - file_url
 - file_size
 - uploaded_by
 - version
 - confidentiality_level
 - created_at
 - deleted_at
-

DocumentVersion

Fields

- id
 - document_id
 - version_number
 - file_url
 - uploaded_by
-

ProjectDocument

Fields

- id
 - project_id
 - document_id
-

TaskDocument

Fields

- id
- task_id
- document_id

Supports documents attached to both projects and tasks simultaneously.

14. Search Domain

SearchIndex

Fields

- id
- entity_type
- entity_id
- searchable_content
- updated_at

Permission checks happen at query time.

15. Analytics Domain

AnalyticsEvent

Fields

- id
- event_type
- event_version
- entity_type
- entity_id
- user_id
- payload
- created_at

AnalyticsSnapshot

Fields

- id
- scope_type
- scope_id
- metric_name
- metric_value
- calculated_at

Scopes

- User
 - Team
 - Department
 - Organization
-

16. Dashboard Domain

Dashboard

Fields

- id
 - owner_id
 - name
 - visibility
 - layout
-

DashboardWidget

Fields

- id
 - dashboard_id
 - widget_type
 - configuration
-

17. Metadata Domain

Tag

Fields

- id
 - name
 - color
 - category
-

CustomField

Fields

- id
 - name
 - field_type
 - entity_type
 - department_id
 - validation_rules
-

CustomFieldValue

Fields

- id
 - custom_field_id
 - entity_type
 - entity_id
 - value
-

18. Automation Domain

AutomationRule

Fields

- id
- name
- trigger
- condition
- action
- is_enabled

Examples

Task Completed → Notify Manager

Leave Approved → Update Attendance

Document Approved → Create Task

19. Activity Domain

Activity

Fields

- id
- actor_id
- entity_type
- entity_id
- action
- metadata
- created_at

Examples

- Task Created
 - Task Assigned
 - Document Approved
 - Project Deleted
-

20. Notification Domain

Notification

Fields

- id
- user_id
- title
- message
- type
- is_read
- created_at

Version 1

- In-App Notifications Only
-

21. Audit Domain

AuditLog

Fields

- id
- user_id
- entity_type
- entity_id
- action
- old_value
- new_value
- ip_address
- timestamp

Rules

- Immutable
 - Never Deleted
-

22. Integration Domain

Integration

Fields

- id
- organization_id
- provider
- status
- configuration

Examples

- Google Workspace
 - Microsoft 365
 - Slack
 - Teams
 - Webhooks
-

Final Status

Requirements : Frozen Architecture : Frozen NFR : Frozen Core ERD : Frozen Permission Matrix : Frozen

This ERD is the official foundation for database design, APIs, workflows, analytics, permissions, dashboards, automation, and future integrations.